1. Solve for x.

   (a) $\sqrt{2x+5}+5=x$

   Solution:

$$\begin{aligned}\sqrt{2x+5}+5=x \implies \sqrt{2x+5}&=x-5\\ \implies 2x+5&=(x-5)^2=x^2-10x+25\\ \implies x^2-12x+20&=0\\ \implies (x-10)(x-2)&=0\\ \implies x&=2,10\end{aligned}$$

   By inspection, only $\boxed{x=10}$ satisfies the original equation.

   (b) $\sqrt{x+2}-\sqrt{x-1}=1$

   Solution:

$$\begin{aligned}&\sqrt{x+2}-\sqrt{x-1}=1 \Rightarrow \sqrt{x+2}=1+\sqrt{x-1}\\ &\Rightarrow x+2=(1+\sqrt{x-1})^2=1+2\sqrt{x-1}+x-1\\ &\Rightarrow x+2=x+2\sqrt{x-1}\\ &\Rightarrow 2=2\sqrt{x-1} \Rightarrow \sqrt{x-1}=1\\ &\Rightarrow x-1=1 \Rightarrow x=2\end{aligned}$$

   Check: $\sqrt{4}-\sqrt{1}=2-1=1$
   So, the solution is $\boxed{x=2}$.

   (c) $\sqrt{x-3}+\sqrt{x-6}=\sqrt{x+2}$

   Solution:

$$\begin{aligned}&\sqrt{x-3}+\sqrt{x-6}=\sqrt{x+2} \implies x-3+2(\sqrt{x-3})(\sqrt{x-6})+x-6=x+2\\ &\implies 2\sqrt{(x-3)(x-6)}=11-x\\ &\implies 4(x-3)(x-6)=(11-x)^2\\ &\implies 4(x^2-9x+18)=121-22x+x^2\\ &\implies 3x^2-14x-49=0\\ &\implies (3x+7)(x-7)=0\\ &\implies x=-\tfrac{7}{3},7\end{aligned}$$

   By inspection, only $\boxed{x=7}$ satisfies the original equation.

(d)

$$\frac{2}{x-1} + \frac{3}{x+2} = 5$$

Solution: Multiply both sides by the LCD $(x-1)(x+2)$

$$\begin{aligned}
&2(x+2) + 3(x-1) = 5(x-1)(x+2) \\
&\implies 2x + 4 + 3x - 3 = 5(x^2 + 5x - 10) \\
&\implies 5x + 1 = 5x^2 + 5x - 10 \\
&\implies 0 = 5x^2 + 5x - 10 - 5x - 1 \\
&\implies 0 = 5x^2 - 11 \\
&\implies x^2 = \frac{11}{5} \\
&\implies x = \pm\sqrt{\frac{11}{5}}
\end{aligned}$$

Check: $x \neq 1, -2$.

(e)

$$\frac{1}{x-2} + \frac{6}{x+2} - \frac{7}{x} = \frac{5}{2}$$

Solution: Multiply both sides by LCD $x(x-2)(x+2)$

$$\begin{aligned}
&x(x+2) + 6x(x-2) - 7(x-2)(x+2) = \frac{5}{2}x(x-2)(x+2) \\
&\implies x^2 + 2x + 6x^2 - 12x - 7(x^2 - 4) = \frac{5}{2}(x^3 - 4x) \\
&\implies x^2 + 2x + 6x^2 - 12x - 7x^2 + 28 = \frac{5}{2}x^3 - 10x \\
&\implies 0 \cdot x^2 + 0 \cdot x + 28 = \frac{5}{2}x^3 \\
&\implies 28 = \frac{5}{2}x^3 \\
&\implies x^3 = \frac{56}{5} \\
&\implies \boxed{x = \sqrt[3]{\frac{56}{5}}}
\end{aligned}$$

(f) $$\frac{\frac{1}{1+x}}{1-\left(\frac{1}{1+x}\right)} = 7$$

Solution:

$$\frac{\frac{1}{1+x}}{\left(\frac{1+x}{1+x}\right)-\left(\frac{1}{1+x}\right)} = 7 \implies \frac{\frac{1}{1+x}}{\frac{x}{1+x}} = 7 \implies \left(\frac{1}{1+x}\right)\left(\frac{1+x}{x}\right) = 7$$

$$\implies \frac{1}{x} = 7 \implies \boxed{x = \frac{1}{7}}$$

2. Simplify the expressions.

(a) $$\frac{4x}{\sqrt{2x}-\sqrt{x}}$$

Solution: Multiply conjugate $(\sqrt{2x}+\sqrt{x})$ on top and bottom

$$\begin{aligned}
\frac{4x}{\sqrt{2x}-\sqrt{x}}\left(\frac{\sqrt{2x}+\sqrt{x}}{\sqrt{2x}+\sqrt{x}}\right) &= \frac{4x(\sqrt{2x}+\sqrt{x})}{2x-x} \\
&= \frac{4x(\sqrt{2x}+\sqrt{x})}{x} \\
&= \boxed{4(\sqrt{2x}+\sqrt{x})}
\end{aligned}$$

(b) $\dfrac{4-\dfrac{1}{x+5}}{\dfrac{7}{x+2}-\dfrac{3}{x+3}}$

Solution:

$$\begin{aligned}
\frac{4-\dfrac{1}{x+5}}{\dfrac{7}{x+2}-\dfrac{3}{x+3}} &= \frac{\dfrac{4(x+5)-1}{x+5}}{\dfrac{7(x+3)-3(x+2)}{(x+2)(x+3)}} \\
&= \left(\frac{4x+19}{x+5}\right)\left(\frac{x^2+5x+6}{4x+15}\right) \\
&= \boxed{\frac{(4x+19)(x^2+5x+6)}{(x+5)(4x+15)}}
\end{aligned}$$

We could obviously continue by multiplying out the numerator and denominator.

3. Provide some choice of numbers x, y, z such that

$$\sqrt{x^2+y^2+z^2} \neq x+y+z.$$

Solution:

Choose $x=y=z=1$ and we have $\sqrt{3}$ vs. 3.